Experimental Design Space and Recycling Formulations Optimization

Summary

Christian Fernandez and Anupam Ajit Deshpande met to evaluate the Design of Experiments (DoE) strategy for polymer recycling formulations across multiple cycles. They reviewed whether to retain the initial 12 formulation parameters across recycling cycles 2 and 3 (Option 1) or generate newly distributed Latin Hypercube Sampling (LHS) points via Latent Variable Gaussian Processes (LVGP) to reduce overall test points (Option 2). Anupam clarified that testing higher recycling cycles for new formulations requires manufacturing and processing filament through cycles 0 and 1 first, which could offset any savings in nominal experiment count. The meeting concluded with Christian agreeing to send an experimental table of 6 and 6 configurations by Friday so Anupam can calculate the required manufacturing time and compare the practical trade-offs.

Important dates

Date	What	Who	Time in recording
Friday, November 06, 2026	Send the DoE configuration table (6 and 6 design points) ("by this Friday")	Christian Fernandez	[16:56]

Next actions

Action	Owner	Due	Time in recording
Send the outline of the discussion to Christian	Anupam Ajit Deshpande	Not stated	[20:21]
Send the DoE formulation table with 6 and 6 configurations for cycles 2 and 3	Christian Fernandez	Friday, November 06, 2026	[16:51], [17:24]
Calculate the experimental manufacturing time for Option 1 vs. Option 2 and email preference	Anupam Ajit Deshpande	Not stated	[14:00], [16:25]
Update the LVGP response surfaces and main effects plots with new uploaded data and reorganize presentation slides	Christian Fernandez	Not stated	[19:19], [20:06]
Discuss design modifications and experimental outline with Doksoo	Christian Fernandez	Not stated	[20:28]

Decisions

- Evaluate Option 1 (retaining the 12 base formulations across cycles 2 and 3) against Option 2 (introducing 6 new formulations for cycle 2 and 6 for cycle 3) based on actual manufacturing and processing workload rather than purely nominal sample counts [13:17], [16:25].

Key points

- **Design Space and Current Status:**
    - The design space covers polymer matrix (PP+PAL, HDPE+PAL, TPU+PAL), PAL content (10–50%), extrusion temperature (180–240 °C), and recycle cycles (0 to 3) [Slide 5, 08:03].
    - Out of 48 planned experiments, 24 have already been completed across cycles 0 and 1 [05:01].
- **Recycling Workflow and Processing Constraints:**
    - The recycling procedure requires extruding pellets into filament, conducting mechanical testing on the filament, 3D printing specimens, performing warpage testing, and shredding the printed parts back into pellets to complete one cycle [06:12].

    - For higher recycling cycles (cycle 2 and cycle 3), starting with a brand new formulation requires running cycles 0 and 1 first [05:32].
    - For Option 1, although 24 tests remain (12 for cycle 2, 12 for cycle 3), the material is already available from previous cycles, so it only requires continuing the cycle chain rather than starting from raw pellets [12:35].

- **Proposed DoE Alternatives (Option 1 vs. Option 2):**
    - **Option 1:** Continue testing the existing 12 formulations across cycles 2 and 3 (12 + 12 = 24 remaining runs) [Slide 11, 09:05].
    - **Option 2:** Reduce new design points using LVGP to distribute points across the space without replicating parameters (e.g., 6 points for cycle 2 and 6 points for cycle 3, totaling 12 runs instead of 24) [Slide 11, 09:46, 15:03].
    - Anupam pointed out that 6 points for cycle 2 requires $6 \times 2 = 12$ cycle runs, and 6 points for cycle 3 requires $6 \times 3 = 18$ cycle runs (totaling 30 cycle operations) if they are completely new formulations [15:43].

- **Preliminary Modeling and Results:**
    - Christian presented preliminary LVGP surface fits for Ultimate Tensile Strength (UTS) across PP and HDPE systems [Slide 7, 17:46].
    - Cycle 1 shows higher average tensile strength than cycle 0 [18:00].
    - Anupam confirmed that mechanical properties often increase initially during early recycling before degrading in later cycles [18:29].

Questions to follow up

- How does the total hands-on manufacturing time for Option 1 compare against Option 2 once Anupam reviews the specific 6 and 6 parameter values? [14:00], [16:25]
- Does the potential insight from non-linear modeling across newly distributed points justify the additional raw filament preparation required for new formulations? [10:16], [13:38]